

BU283

Financial Management I

MIDTERM NOTES OT ALIMOV FALL 2026

CH.3
Time Value of Money

A dollar today is worth more than a dollar later, because today's dollar can be invested and earn interest. Every formula in this chapter is one equation solved for a different unknown.

FORMULA The master equation

$$FV_n = PV_0\,(1+i)^n$$

Know any 3 of FV, PV, i, n and you can solve for the 4th.

3.1 Simple Interest and Timelines

DEF Simple interest

Interest earned on the original principal only. Interest never earns interest.

DEF Compound interest

Interest earned on principal plus all interest already accumulated.

FORMULA Simple interest

$$\text{Interest} = P \times i$$

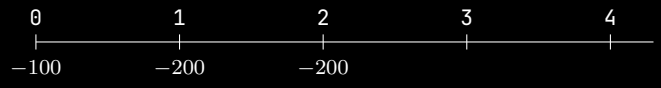
Shown in the course as what NOT to do.

EXAMPLE $100 for one year at 5%

Interest $= 100 \times 0.05 = \$5$

DEF Timeline

Time 0 (today) is the far left, future periods move right. Time 1 is the END of period 1 and the START of period 2.



DEF Inflow vs outflow

Inflow = cash you RECEIVE, positive (+). Outflow = cash you SPEND, negative (−).

WATCH OUT

On the calculator, PV and FV need opposite signs or you get an error solving for i or n.

3.2 Future Value of a Sum

DEF Future value (FV)

What money invested today grows to by a future date.

DEF Compounding

Moving cash flows RIGHT on the timeline, from today into the future.

FORMULA One period

$$FV_1 = PV_0\,(1+i)$$

FORMULA Multiple periods (Eq. 3.1)

$$FV_n = PV_0\,(1+i)^n$$

$(1+i)^n$ is the future value interest factor (FVIF).

EXAMPLE $1,000 at 5% for two years

Year 1: $1000(1.05) = \$1,050$
Year 2: $1000(1.05)^2 = \$1,102.50$
Simple interest gives $1,100. The extra $2.50 is interest on interest.

EXAMPLE $1,000 at 10% for 40 years

$FV_{40} = 1000(1.10)^{40} = \$45,259.26$

FORMULA Interest earned

$$\text{Interest earned} = FV - PV$$

WATCH OUT

Slide 14 says $35,259.26. Correct is $45,259.26 - 1,000 = \$44,259.26$.

DEF Rules of thumb

FV rises when i rises or n rises. Compounding applies to any growth: population, sales, dividends.

DEF Mixed stream

Different cash flows at different times. Compound each one to the end date separately, then add.

EXAMPLE Ex 3.3, FV of a mixed stream

$100 (end yr 1), $200 (end yr 2), $300 (end yr 3) invested at 10%. FV at end of year 6?

$$100(1.10)^5 = 161.05$$
$$200(1.10)^4 = 292.82$$
$$300(1.10)^3 = 399.30$$
$$FV_6 = \$853.17$$

WATCH OUT

The exponent is how many periods the cash flow GROWS, not the year it arrives. $100 arrives at year 1 and grows 5 years.

3.2b Non-Annual Compounding

DEF Compounding frequency, m

Times compounded per year. Annual 1, semiannual 2, quarterly 4, monthly 12, daily 365.

FORMULA FV with m periods per year (Eq. 3.3)

$$FV_n = PV_0\left(1 + \frac{i}{m}\right)^{mn}$$

Divide the rate by m. Multiply the years by m.

EXAMPLE $1,500 for 20 years at 8% compounded quarterly

$$FV = 1500\left(1 + \tfrac{0.08}{4}\right)^{80} = 1500(1.02)^{80} = \$7,313.16$$

DEF Effect of frequency

More compounding periods means higher FV, but each extra increase gets smaller.

3.3 Effective Interest Rate

DEF **Quoted (nominal) rate**

The stated annual rate i, before compounding. Also called APR.

DEF **Effective interest rate (EIR)**

The rate you REALLY earn per year after compounding. Use it to compare rates with different compounding.

FORMULA **Effective interest rate**

$$EIR = \left(1 + \frac{i}{m}\right)^{m} - 1$$

i = quoted annual rate, m = compounding periods per year

EXAMPLE **Ex 3.5, 12% compounded monthly**

$$EIR = (1.01)^{12} - 1 = 12.68\%$$

DEF **Key fact**

EIR $\geq$ quoted rate. Equal only when $m = 1$.

3.4 Solving for i and n

FORMULA **Interest rate (Eq. 3.5)**

$$i = \left(\frac{FV_n}{PV_0}\right)^{1/n} - 1$$

EXAMPLE **$1,000 grows to $2,000 in 5 years**

$$i = 2^{1/5} - 1 = 14.87\%$$

FORMULA **Number of periods**

$$n = \frac{\ln(FV_n/PV_0)}{\ln(1+i)}$$

Excel: =NPER(rate, pmt, pv, fv)

EXAMPLE Double $1,000 at 12%

$$n = \frac{\ln 2}{\ln 1.12} = 6.12 \text{ years}$$

Calculator: PV -1000, FV 2000, I/Y 12, PMT 0, CPT N.

DEF Rule of 72

Years to double $\approx 72/\text{rate}$. At 12%: $72/12 = 6$. Good sanity check.

3.5 Present Value of a Sum

DEF Present value (PV)

What a future cash flow is worth today.

DEF Discounting

Computing PV. Moving cash flows LEFT on the timeline. The rate used is the discount rate.

FORMULA Present value (Eq. 3.6)

$$PV_0 = \frac{FV_n}{(1+i)^n}$$

$1/(1+i)^n$ is the present value interest factor (PVIF).

EXAMPLE $100 in 10 years at 5%

$$PV = \frac{100}{1.05^{10}} = \$61.39$$

DEF PV insights

PVIF is the inverse of FVIF. If $i = 0$ then PV = FV. Higher rate or longer wait means lower PV.

DEF Intuition

PV of $100 in a year at 10% is $90.91. You are indifferent between the two, because $90.91 invested at 10% becomes exactly $100.

EXAMPLE PV of a mixed stream at 10%

$100, $200, $300, $400 at end of years 1 to 4.

$$100/1.10^{1} = 90.91$$
$$200/1.10^{2} = 165.29$$
$$300/1.10^{3} = 225.39$$
$$400/1.10^{4} = 273.21$$
$$PV_0 = \$754.80$$

FORMULA PV with m periods per year (Eq. 3.7)

$$PV_0 = \frac{FV_n}{\left(1 + \frac{i}{m}\right)^{mn}}$$

EXAMPLE $5,000 in 3 years, 6% compounded monthly

$$PV = \frac{5000}{1.005^{36}} = \$4,178.22$$

3.6 Discounted Cash Flow

DEF DCF

The value of ANY asset is the present value of the cash flows it generates.

DEF DCF in 3 steps

1. Identify the asset’s cash flows.
2. Pick a discount rate that pays for the asset’s risk.
3. Discount each cash flow at that rate and add.

calc BA II Plus

DEF Keys

N periods, I/Y rate in % (type 12, not 0.12), PV, PMT (0 for a single sum), FV. Press CPT then the unknown.

WATCH OUT

Clear first with 2ND, CLR TVM. Keep P/Y = 1 and adjust N and I/Y yourself for non-annual compounding.

check Quick Questions

Q1

Why does compound interest beat simple interest?

A. Each period’s interest joins the balance and earns interest itself.

Q2

If the discount rate rises, what happens to PV?

A. PV falls. Less money today is needed to grow into the same future amount.

Q3

Bank A: 10% compounded annually. Bank B: 9.6% compounded monthly. Which pays more?

A. Bank B, barely. EIR $= (1 + 0.096/12)^{12} - 1 = 10.03\%$.

Q4

At what rate does PV equal FV?

A. $i = 0$, since $(1 + 0)^n = 1$.

Q5

Why must PV and FV have opposite signs on the calculator?

A. One is money out, the other is money back. Same signs and no rate can balance it.